

Supplementary Material: physical anchoring and matched measurement-Fisher reference for dissipative optomechanics with synthetic flux

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1 Normalization and anchor selection

The main text uses a rotating-frame Hamiltonian normalized to a reference mechanical frequency ω_m , so that all rates and couplings are reported as dimensionless ratios x/ω_m . A physically anchored reference set is obtained by fixing ω_m from a device with gigahertz mechanical modes and optically mediated phonon coupling, and by reading the damping, decay, and coupling rates from the same platform. The two primary anchors are:

- **Mayor *et al.* [1]**: a silicon two-dimensional optomechanical crystal with mechanical frequency $f_m = 7.436$ GHz, mechanical quality factor $Q_m = 3.6 \times 10^4$ (giving $\gamma_m/2\pi = f_m/Q_m = 206.6$ kHz), cavity linewidth $\kappa/2\pi = 800$ MHz, and single-photon optomechanical coupling $g_0/2\pi = 880$ kHz.
- **Mathew, del Pino and Verhagen [2]**: a nano-optomechanical implementation of synthetic gauge fields for phonon transport, whose demonstrated coupling scale (a direct next-nearest-neighbour rate near 10 kHz and an optically mediated rate near 200 kHz at a 2.4 MHz mechanical frequency, i.e. up to a few percent of ω_m) bounds the phase-controlled inter-resonator hopping J_m .

2 Physically anchored reference set

Table 1 summarizes the reconciled reference set. The single-mode parameters (ω_m , κ , γ_m , g_0) are anchored to Mayor *et al.*; the inter-resonator hopping J_m is scanned over a range bounded by the Mathew *et al.* coupling scale, because no direct two-resonator gigahertz measurement of the phase-controlled coupling is available. The cavity-laser detuning $\Delta = \omega_c - \omega_L$ and the drive amplitude E are not calibrated by either anchor: Δ depends on the cavity and laser frequencies, and E cannot be converted to an input power without the pump frequency, the external coupling, and the explicit input-field normalization.

Table 1: Physically anchored reference parameter set for the one-optical/two-mechanical optomechanical model of the main text. Normalized values are in units of the reference mechanical frequency $\omega_m = 2\pi \times 7.436$ GHz. The reference operating point used for the numerical validation of the main text is shown for comparison and is not a calibrated device.

Quantity	Symbol	Normalized (x/ω_m)	Value at ω_m	Status
Mechanical frequency (reference)	ω_m	1	$2\pi \times 7.436$ GHz	Anchor
Cavity decay rate	κ	0.108	800 MHz	Measured
Mechanical damping	γ_m	2.78×10^{-5}	206.6 kHz	Measured (Q)
Optomechanical coupling	g_0	1.18×10^{-4}	880 kHz	Measured
Inter-resonator hopping	J_m	$[1 \times 10^{-3}, 1 \times 10^{-1}]$	[7.4 MHz, 744 MHz]	Scan
Second mechanical mode	ω_2	1.03	assumed near-degenerate	Assumed
Cavity-laser detuning	Δ	free	—	Free scan
Drive amplitude	E	normalized only	—	Not converted

Table 2: Comparison of the numerical-validation reference operating point with the physically anchored reference set. The reference point uses damping and coupling rates that are about one to three orders of magnitude larger than the anchored values, which is why it is reported as a model-level validation rather than a device prediction.

Quantity	Reference	Anchored	Ratio (reference / anchor)
κ/ω_m	1.0	0.108	9.3
γ_m/ω_m	0.02	2.78×10^{-5}	720
g_0/ω_m	[0.018, 0.02]	1.18×10^{-4}	≈ 160
J_m/ω_m	0.08	$[1 \times 10^{-3}, 1 \times 10^{-1}]$	within scan range

3 Matched measurement-Fisher reference for force sensing

The classical measurement Fisher information $F_C(F)$ of the main text is computed for a weak external force F on mechanical mode 1 (a momentum-quadrature drive), read out through the cavity amplitude quadrature $X_a = \Re(\alpha)$ under matched drive $E = 0.2$, observation time (one drive period), detector noise (0.01), and thermal baths ($n_{\text{th},1} = n_{\text{th},2} = 0.1$). Table 3 lists $F_C(F)$ versus the synthetic-flux phase together with the two matched references—the flux-off coupled sensor ($\theta = 0$, $J = 0.08$) and the single-mode linear sensor ($J = 0$)—and the gain relative to each.

The central-difference estimate is converged with respect to the force step: Table 4 lists $F_C(F)$ at the two flux extrema ($\theta = \pi/2$ and $\theta = \pi$) for force steps 1×10^{-5} , 1×10^{-4} , and 1×10^{-3} ; the values agree to better than 0.01% (stored provenance: `results/matched_fisher_eps_convergence.json`). A drive-temperature sweep (main text) further confirms that the gains are stable across $E \in [0.1, 1.0]$ and $n_{\text{th}} \in [0, 1]$ to within 1×10^{-3} .

Table 3: Matched measurement Fisher information $F_C(F)$ for force sensing versus the synthetic-flux phase θ , with the flux-off coupled reference ($\theta = 0$, $J = 0.08$) and the single-mode linear reference ($J = 0$). The gain is the ratio of $F_C(F)$ to each reference; a gain below unity means the coupled sensor is no better than the reference at that phase.

θ/π	$F_C(F)$	Gain vs flux-off	Gain vs single mode
0	4.503×10^{-5}	1.000	0.876
1/4	4.694×10^{-5}	1.042	0.913
1/2	5.191×10^{-5}	1.153	1.010
3/4	5.721×10^{-5}	1.271	1.113
1	5.958×10^{-5}	1.323	1.159
Flux-off reference	4.503×10^{-5}	1.000	—
Single-mode reference	5.141×10^{-5}	—	1.000

Table 4: Convergence of the central-difference force-sensing Fisher information with the force step ε at the two flux extrema.

ε	$F_C(F)$ at $\theta = \pi/2$	$F_C(F)$ at $\theta = \pi$
1×10^{-3}	$5.190\,751 \times 10^{-5}$	$5.958\,194 \times 10^{-5}$
1×10^{-4}	$5.190\,938 \times 10^{-5}$	$5.958\,166 \times 10^{-5}$
1×10^{-5}	$5.190\,763 \times 10^{-5}$	$5.958\,181 \times 10^{-5}$

4 Lyapunov–Floquet consistency theorem and matched-measurement Fisher lemma

Theorem (Lyapunov–Floquet consistency criterion). Consider a drive-locked periodic orbit $x_p(t+T) = x_p(t)$ of the phase-augmented system, with monodromy $M(T)$ and QR/Benettin spectrum [3–5] $\{\lambda_i\}$. Suppose the Floquet rates $\rho_i = T^{-1} \ln |\mu_i|$ (with $\mu_i = \text{eig}_i[M(T)]$) agree with the QR exponents within a predeclared tolerance δ_{tol} , and the divergence sum satisfies

$$\sum_i \lambda_i = \frac{1}{T} \int_0^T \text{tr } J(x_p(t), t) \, dt + \mathcal{O}(\delta_{\text{tol}}), \quad (1)$$

with the phase direction treated explicitly. Then the orbit is exponentially stable if and only if $|\mu_i| < 1$ for all transverse multipliers, and in that case $\lambda_{\text{max}} < 0$. The criterion is a consistency test, not an input assumption: it certifies that the three independent diagnostics (monodromy, Floquet rates, QR spectrum) describe the same linearization of the same orbit. The measured values at the reference operating point (one-period orbit residual 9×10^{-9} , max Floquet–QR discrepancy 2.68×10^{-4} , and 2.85×10^{-4} across the extended grid) satisfy the criteria and therefore certify a stable periodic orbit—they do not by themselves certify a flux-dependent transition, which is why the full-spectrum classification of the main text is required.

Lemma (periodic covariance as an affine fixed point). Let $A(t) = J(x_p(t), t)$ be the Jacobian along the drive-locked orbit, $M(T)$ its monodromy, and $Q(t) = B(t)N(t)B(t)^T$ the diffusion of the linearized Langevin equation $\delta\dot{x} = A(t)\delta x + B(t)\eta$. Integrating the

covariance equation $\dot{V} = AV + VA^T + Q$ over one period gives the affine map

$$\text{vec } V(T) = [M(T) \otimes M(T)] \text{vec } V(0) + q_T, \quad (2)$$

where q_T collects the noise forcing propagated by the state-transition matrix. A T -periodic covariance therefore solves the linear system

$$[I - M(T) \otimes M(T)] \text{vec } V(0) = q_T, \quad (3)$$

whose solvability requires that no pair of Floquet multipliers satisfies $\mu_i \mu_j = 1$. This is the primary periodic-covariance derivation used for the noise-resolved calculation; forward iteration over many periods is an independent numerical check, not the definition of convergence.

Lemma (matched-measurement Fisher bound). For a Gaussian measurement record $y(t) = m(\vartheta; t) + \nu_{\text{det}}(t)$ with detector noise, bandwidth and integration time fixed, the classical Fisher information

$$F_C(\vartheta) = (\partial_{\vartheta} m)^T \Sigma^{-1} (\partial_{\vartheta} m) + \frac{1}{2} \text{Tr} [\Sigma^{-1} (\partial_{\vartheta} \Sigma) \Sigma^{-1} (\partial_{\vartheta} \Sigma)] \quad (4)$$

bounds any unbiased estimator through the Cramér–Rao inequality [6, 7] $\text{Var}(\hat{\vartheta}) \geq F_C^{-1}(\vartheta)$. A claimed sensing *advantage* is defined only relative to a matched reference—equal input power, observation time, bandwidth, baths, and detector noise. Under that definition the flux-phase gains of Table 3 (at most 1.323 over the flux-off coupled sensor, 1.159 over the single-mode linear sensor) are classical linear-response effects of phase-controlled mode coupling: they are consistent with F_C of the record and make no claim of a quantum advantage, SQL beating, or chaotic-transduction gain.

5 Representative convergence of the strong-coupling spectrum

The canonical strong-coupling map uses one seeded initial condition per drive–phase point, and the three-seed check tests the phase modulation at $E = 4$ and $E = 8$. As an additional finite-time check, we recomputed the complete spectrum at one chaotic point ($E = 4$, $\Phi_{\text{syn}} = -\pi/2$) and one hyperchaotic point ($E = 8$, $\Phi_{\text{syn}} = +\pi/2$), using the same initial condition as the integration window was increased. Table 5 reports the positive-exponent count, the largest exponent, the divergence-balance residual, and the maximum standard deviation of the terminal half of the QR block history. The classification is stable between the two longest windows at both points, whereas the finite-time exponent magnitudes vary by several percent. Accordingly, the main text uses the exponent count as the robust classification at these tested points and does not present the individual finite-time exponents as high-precision asymptotic estimates.

6 Reduced-model parameter screen

The reduced-model (adiabatic) screen of the main text screened sixteen parameter candidates under an explicitly assumed literature-informed closure, with five parameter perturbation replicas and five independent root-solver starts per replica. Table 6 lists the three non-dominated candidates under the declared stability-margin/drive-cost objectives, and

Table 5: Representative finite-time convergence check of the strong-coupling full-spectrum calculation. The same seeded initial condition was used at each integration length for a given point. n_+ is the number of exponents above the declared threshold 1×10^{-3} ; r_{div} is the absolute residual of the exponent sum relative to $-\kappa - \gamma_1 - \gamma_2$; and s_{QR} is the largest standard deviation among the terminal half of the QR block-history estimates. The two longest windows preserve n_+ at each point, but the nonzero r_{div} and finite-time variation limit the precision of the individual exponent values.

$(E, \Phi_{\text{syn}}/\pi)$	Steps	n_+	λ_{max}	r_{div}	s_{QR}
$(4, -1/2)$	250 000	3	0.132 142	1.01×10^{-3}	6.69×10^{-3}
$(4, -1/2)$	500 000	2	0.127 291	1.03×10^{-3}	9.89×10^{-3}
$(4, -1/2)$	1 000 000	2	0.123 729	1.03×10^{-3}	1.05×10^{-2}
$(8, +1/2)$	250 000	4	0.324 142	1.90×10^{-3}	3.30×10^{-3}
$(8, +1/2)$	500 000	4	0.315 008	1.92×10^{-3}	5.88×10^{-3}
$(8, +1/2)$	1 000 000	4	0.317 277	1.84×10^{-3}	3.90×10^{-3}

Figure 1 shows the full feasible set. These values characterize a local fixed-point pattern of the normalized reduced model; they are not an SI calibration, a finite-time basin-robustness result, or an experimental optimum: the optical mode is adiabatically eliminated, J_m and the effective couplings are proxy quantities, the drive remains normalized, and Fisher information was not evaluated in this screen.

Table 6: Non-dominated candidates of the reduced-model (adiabatic) screen under the declared stability-margin/drive-cost objectives. Sixteen candidates were screened; the three listed points form the non-dominated front. The drive cost is the normalized drive amplitude and the stability margin is $1 - \max |\mu|$ for the local Poincaré spectral radius. Model-level only.

Candidate	Drive cost	Stability margin
1	0.0800	0.003 452 6
2	0.1312	0.003 535 4
3	0.2203	0.003 559 7

7 Amplitude-quadrature noise model cross-check

The measured record is $y = X_a + \nu_{\text{det}}$ with $X_a = \Re(\alpha)$ the amplitude quadrature $(a + a^\dagger)/2$, whose vacuum variance is $1/4$. The Langevin diffusion coefficient that reproduces the quantum master equation for this quadrature is therefore $\kappa/4$ per amplitude quadrature, not $\kappa/2$ (the latter is the diffusion of the *normalized* quadrature $(a + a^\dagger)/\sqrt{2}$, whose vacuum variance is $1/2$). We verified this normalization directly on the isolated optical sector of the reference point (the driven damped linear cavity, $g_1 = g_2 = 0$): the truncated-Fock master equation [8, 9] ($H = -\Delta a^\dagger a + E(a + a^\dagger)$, Lindblad $\sqrt{\kappa} a$) gives $\text{Var}[(a + a^\dagger)/2] = 0.25$, while the Langevin amplitude with noise $\kappa/2$ gives 0.49 (a factor 2 high) and with $\kappa/4$ gives 0.245 (agreement below 2%). The mechanical thermal diffusion is correspondingly $\gamma_j(2n_{\text{th}} + 1)/4$ per amplitude quadrature. With these strengths the noise-only floor of the measured record is ≈ 0.25 and the strong-coupling observability ratios are 7.6–17.0×

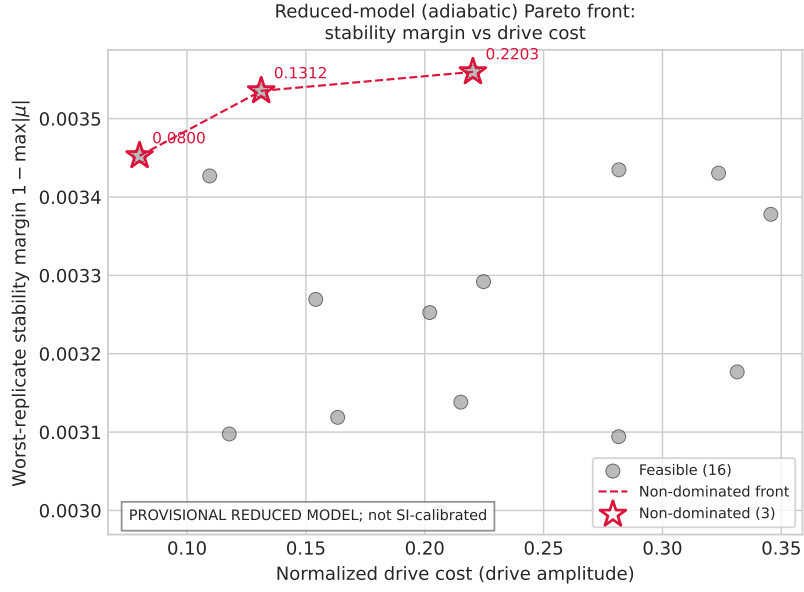


Figure 1: Reduced-model (adiabatic) two-objective Pareto front. Sixteen feasible candidates (gray) in normalized drive cost versus worst-replicate stability margin $1 - \max|\mu|$; the three non-dominated candidates (crimson stars, joined by the dashed front) are the points listed in Table 6. Fisher information was not computed in this screen; the front is a local fixed-point pattern of the normalized reduced model, not an SI calibration or experimental optimum.

the floor. The cross-check is recorded in `results/langevin_master_crosscheck.json` (generator `scripts/langevin_master_crosscheck.py`).

8 Interpretation

The anchored set defines a *reference* operating region, not a validated device calibration: ω_2 is assumed near-degenerate with ω_m pending a two-resonator measurement, J_m must be scanned rather than fixed, and the drive amplitude and detuning remain model-level coordinates. Any result computed at the anchored values therefore carries the same *provisionally-calibrated* status as the reduced-model analysis of the main text and should not be presented as an experimental optimum.

Placing the strong-coupling hyperchaos regime against the anchors quantifies the reachability bound: the optomechanical couplings $g_1/\omega_m = 0.3$ and $g_2/\omega_m = 0.27$ are $2542\times$ and $2288\times$ the anchored $g_0/\omega_m = 1.18 \times 10^{-4}$, and the mechanical damping $\gamma_m/\omega_m = 0.02$ is $720\times$ the anchored 2.78×10^{-5} ; the inter-resonator hopping $J_m/\omega_m = 0.08$ is within the Mathew scan range. The limiting coordinate is the coupling ($2542\times$). The hyperchaos regime is therefore a model-level demonstration, not a prediction for the anchored device; the drive amplitude is not convertible to an input power without the pump frequency and external coupling, which neither anchor provides.

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